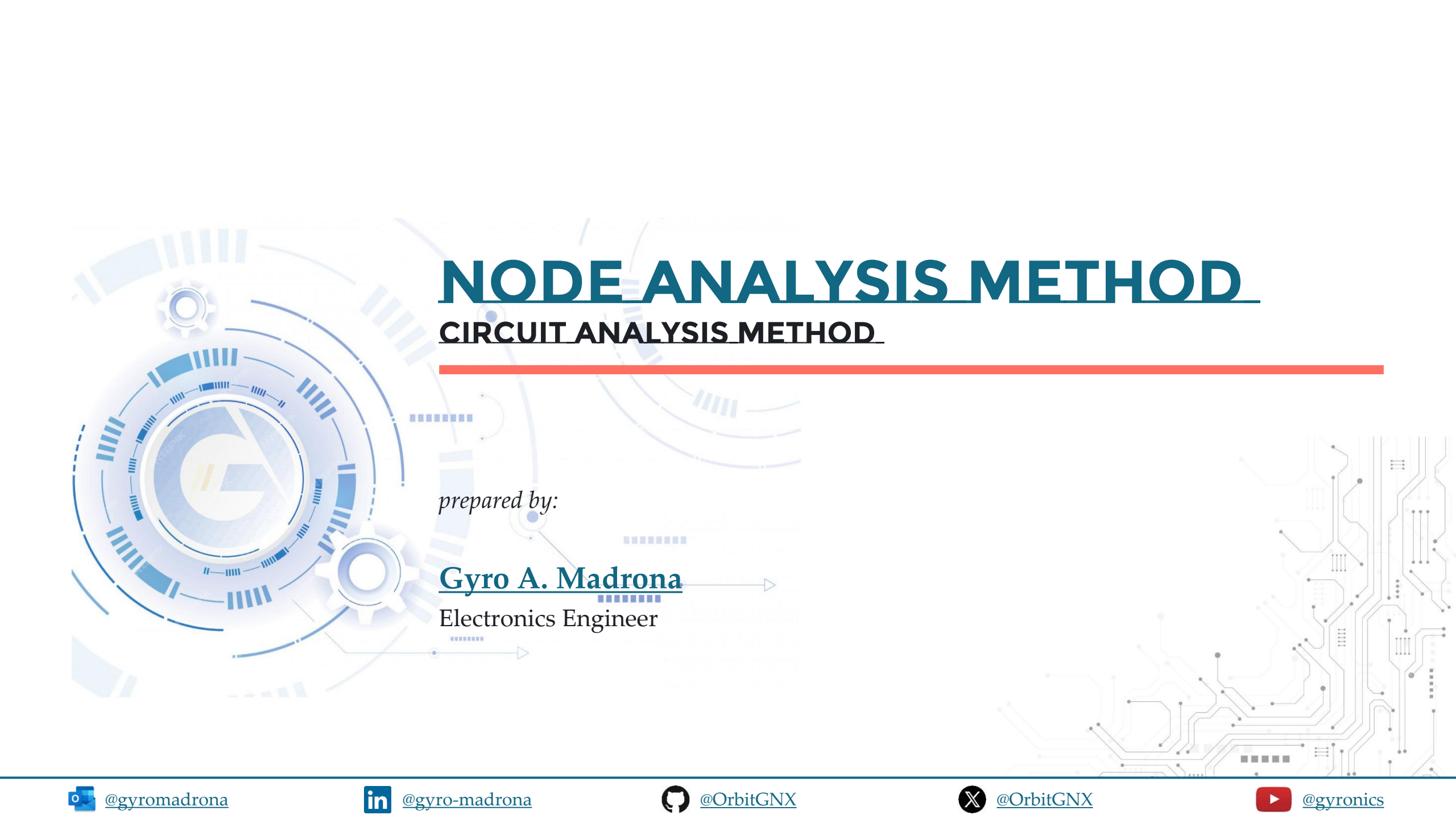




NODE ANALYSIS METHOD

CIRCUIT ANALYSIS METHOD

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TOPIC OUTLINE

Node Voltage

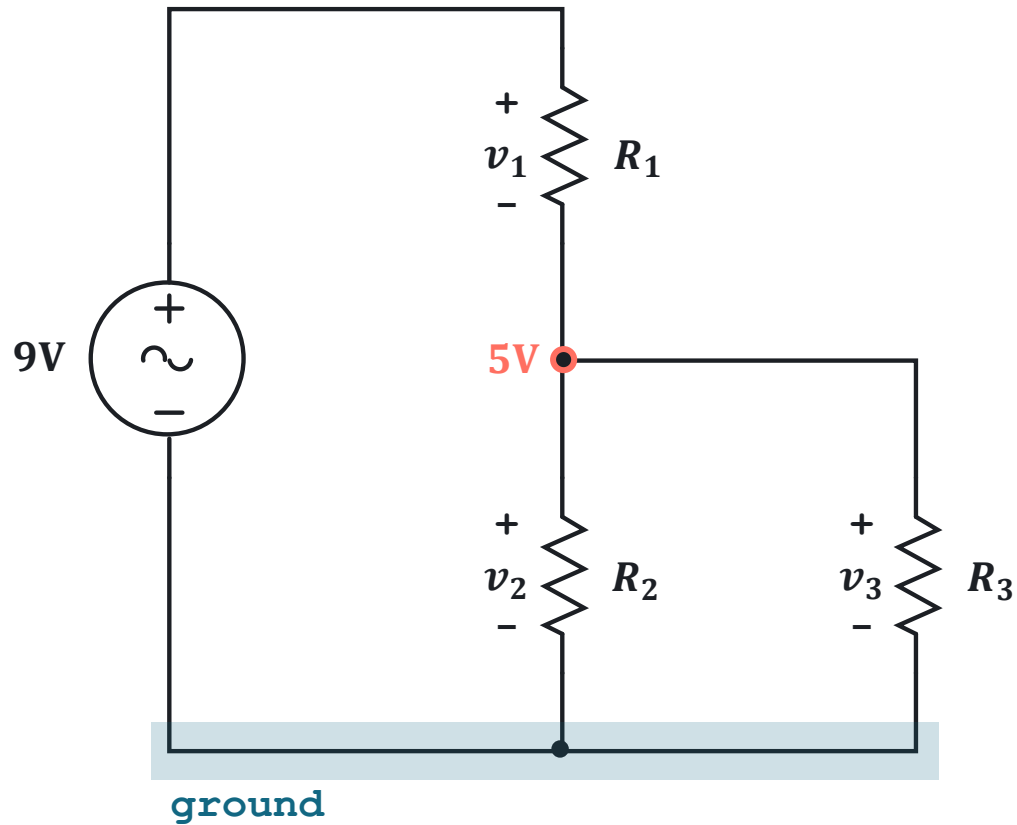
Node Analysis Method



NODE ANALYSIS METHOD



NODE VOLTAGE



Node Voltage is the electrical potential difference between a given node and a reference node (ground) in a circuit.

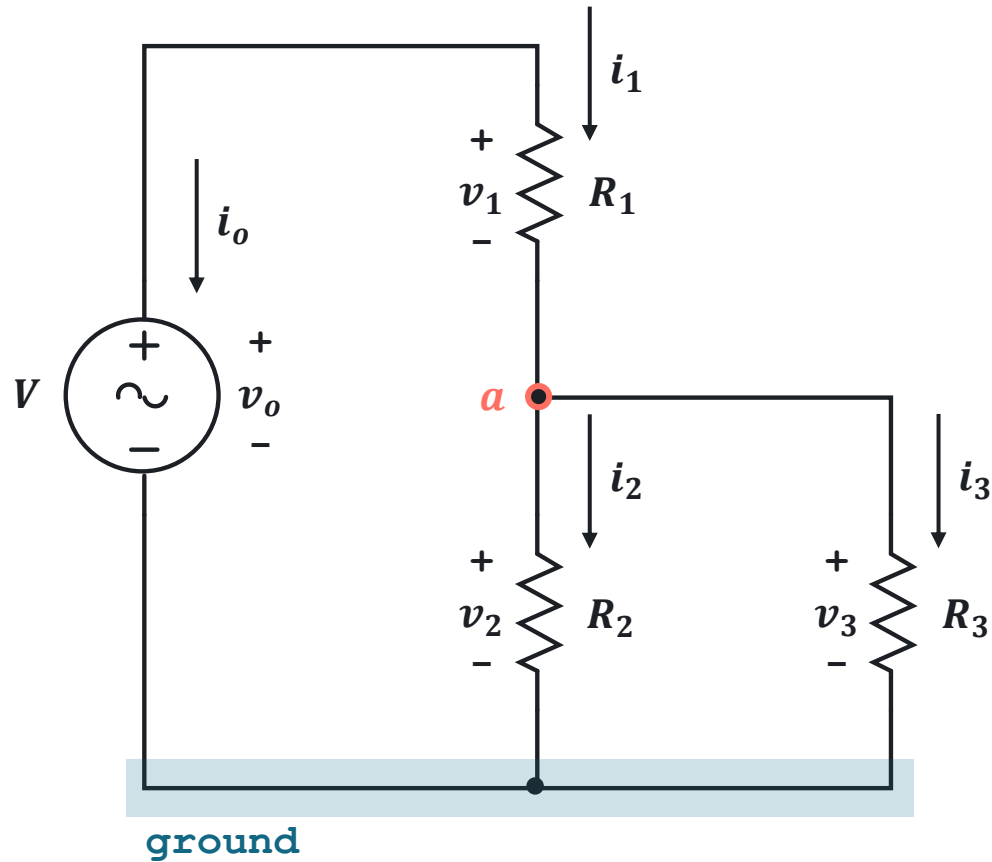
$$v_1 = 9V - 5V$$

$$v_2 = 5V - 0$$

$$v_3 = 5V - 0$$



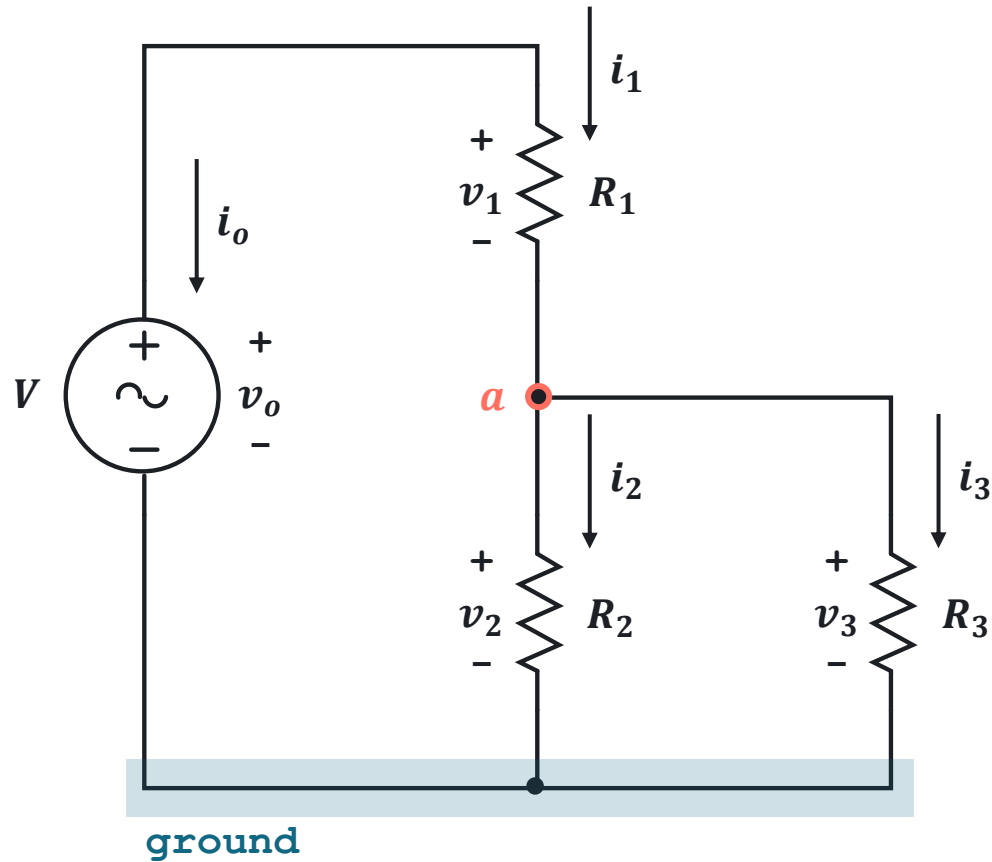
NODE ANALYSIS METHOD



The node analysis method is based on Kirchhoff's Current Law (KCL), which is implicitly applied to establish voltage-current relationships forming a system of equations to be solved for the unknown node voltages.



NODE ANALYSIS METHOD



KCL@a

$$-i_1 + i_2 + i_3 = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

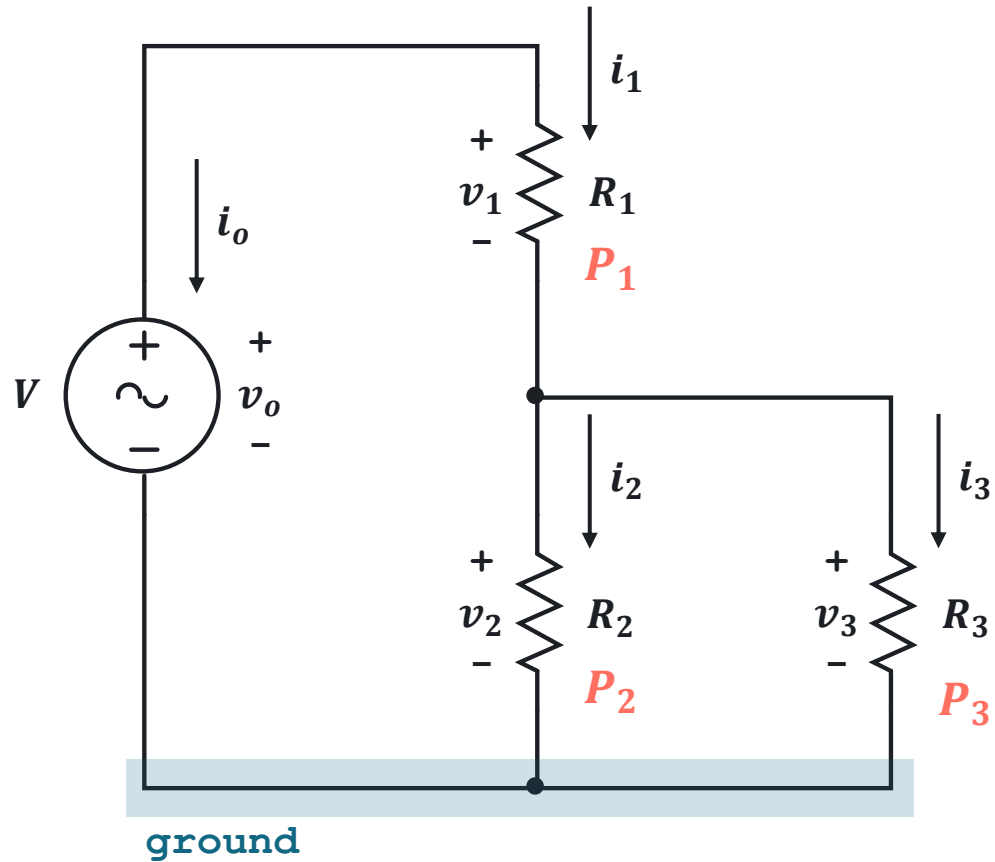
$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_a - 0}{R_3} = 0$$

$$-v_o G_1 + v_a G_1 + v_a G_2 + v_a G_3 = 0$$

$$v_a(G_1 + G_2 + G_3) = v_o G_1$$



ELECTRICAL POWER



Electrical power refers to the rate at which electrical energy is transferred or converted per unit time.

Formulas

$$P = vi$$

$$P = i^2 R$$

$$P = \frac{v^2}{R}$$

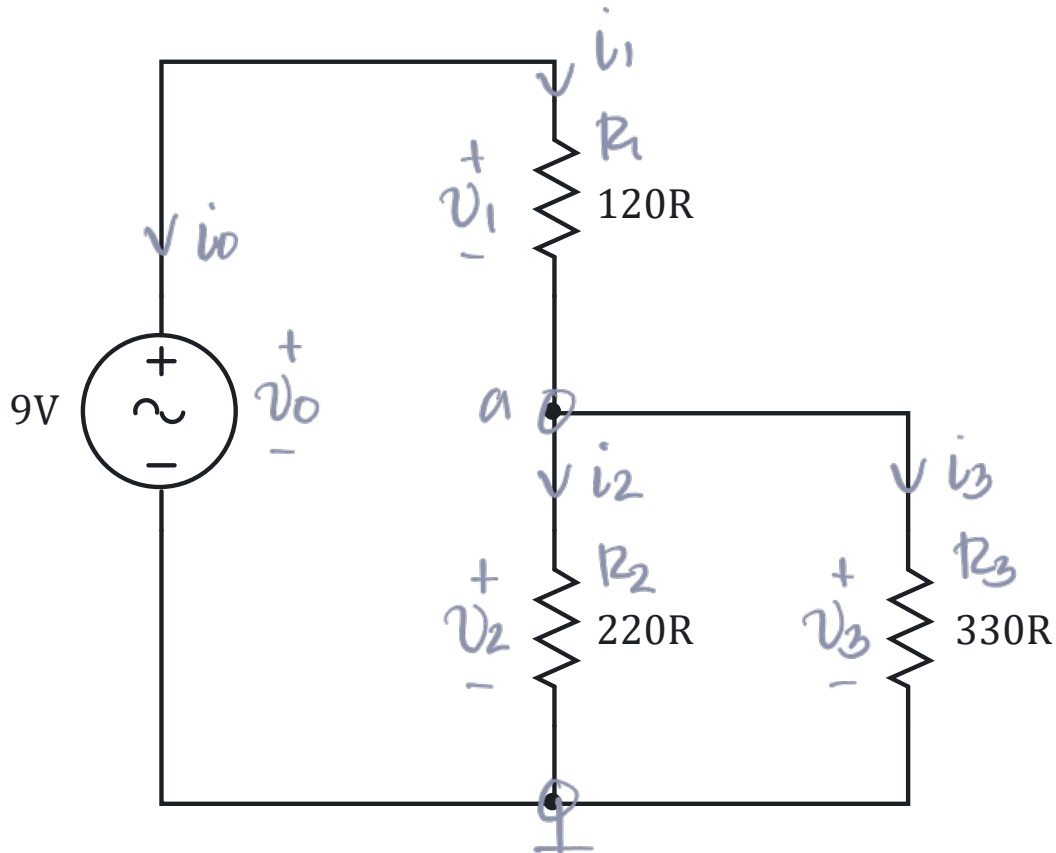
$$P_{total} = P_1 + P_2 + P_3 + \cdots P_n$$

unit: Watt (W)



EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

KCL @ a

$$-i_1 + i_2 + i_3 = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

$$-\frac{v_0 - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_a - 0}{R_3} = 0$$

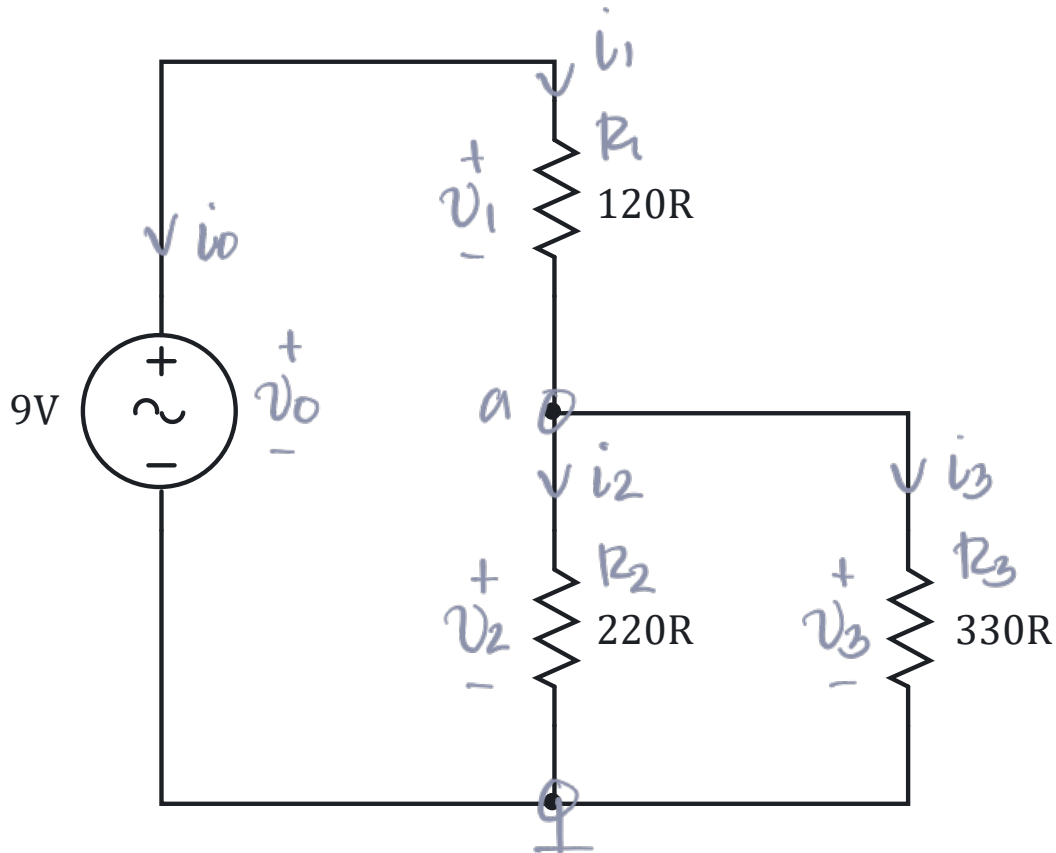
$$-\frac{v_0}{R_1} + \frac{v_a}{R_1} + \frac{v_a}{R_2} + \frac{v_a}{R_3} = 0$$

$$-v_0 G_1 + v_a G_1 + v_a G_2 + v_a G_3 = 0$$

$$v_a (G_1 + G_2 + G_3) = v_0 G_1$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$V_a = \frac{V_0 G_1}{G_1 + G_2 + G_3}$$

$$V_a = \frac{9 \left(\frac{1}{120} \right)}{\frac{1}{120} + \frac{1}{220} + \frac{1}{330}}$$

$$\underline{V_a = 4.71 \text{ V}}$$

$$\boxed{V_2 = 4.71 \text{ V}}$$

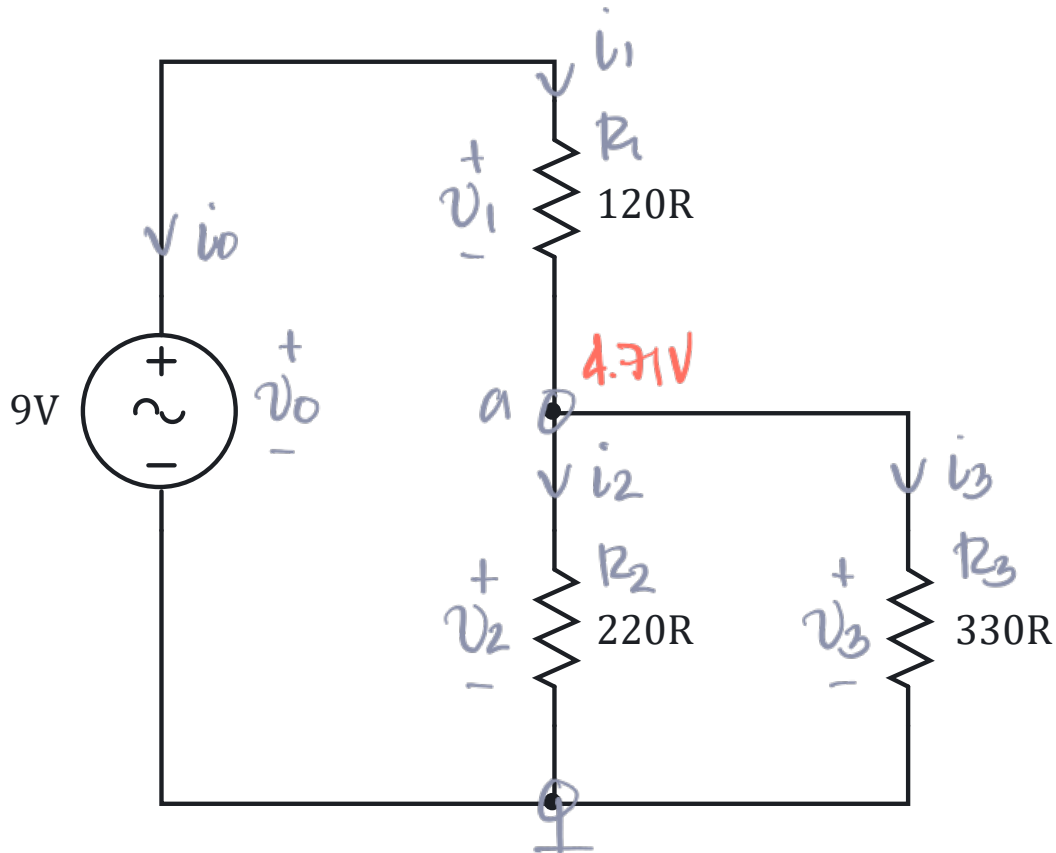
ans

$$\boxed{V_3 = 4.71 \text{ V}}$$

ans

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$v_1 = v_0 - v_a$$

$$v_1 = 9 - 4.71$$

$$v_1 = 4.29 \text{ V}$$

ans

$$i_1 = \frac{v_1}{R_1}$$

$$i_1 = \frac{4.29}{120}$$

$$i_1 = 35.75 \text{ mA}$$

ans

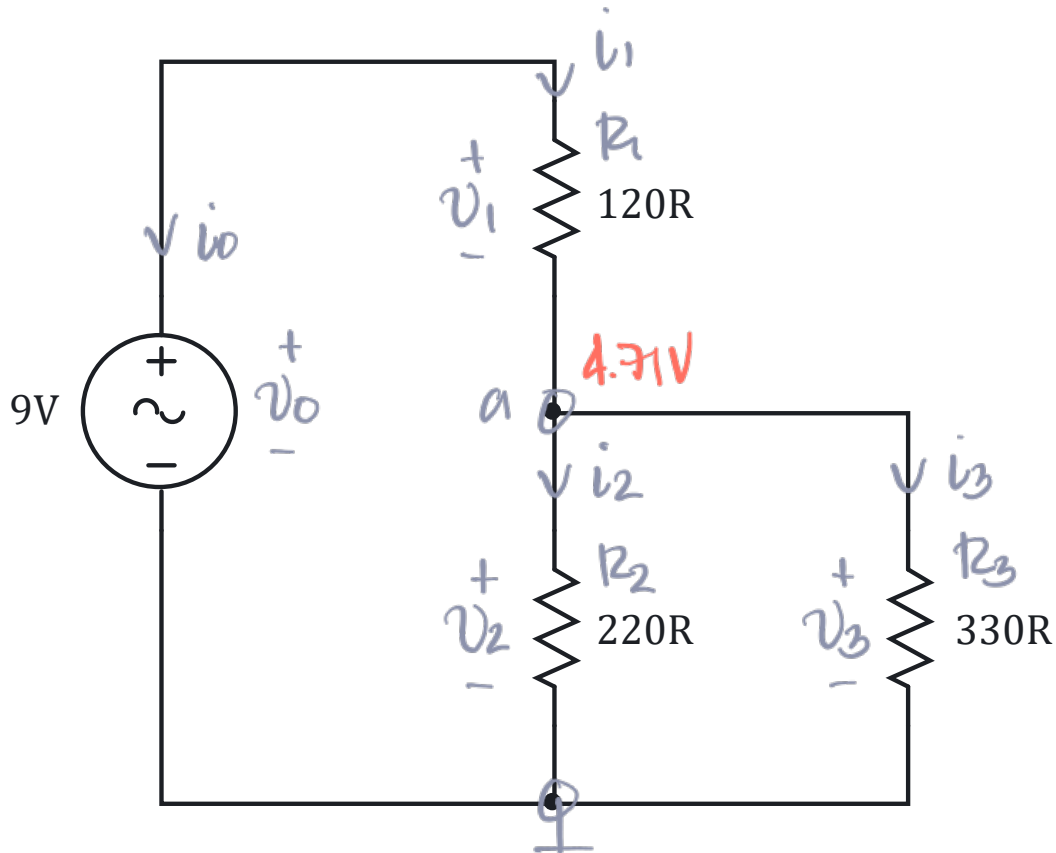
$$P_1 = i_1 v_1$$

$$P_1 = 35.75 \text{ m}(4.29)$$

$$P_1 = 153.37 \text{ mW}$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$i_2 = \frac{v_2}{R_2}$$

$$i_2 = \frac{4.71}{220}$$

$$i_2 = 21.41 \text{ mA}$$

ans

$$P_2 = i_2 v_2$$

$$P_2 = 21.41 \text{ m}(4.71)$$

$$P_2 = 100.84 \text{ mW}$$

$$i_3 = \frac{v_3}{R_3}$$

$$i_3 = \frac{4.71}{330}$$

$$i_3 = 14.27 \text{ mA}$$

ans

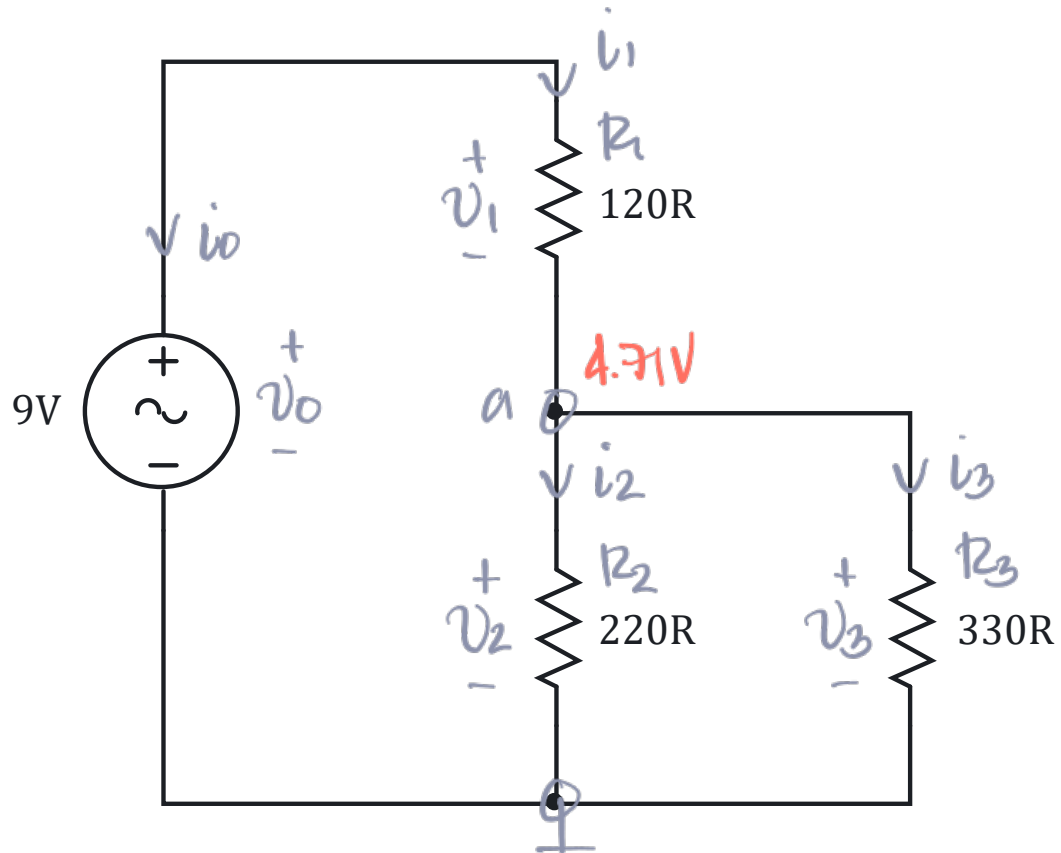
$$P_3 = i_3 v_3$$

$$P_3 = 14.27 \text{ m}(4.71)$$

$$P_3 = 67.21 \text{ mW}$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$P_0 = P_1 + P_2 + P_3$$

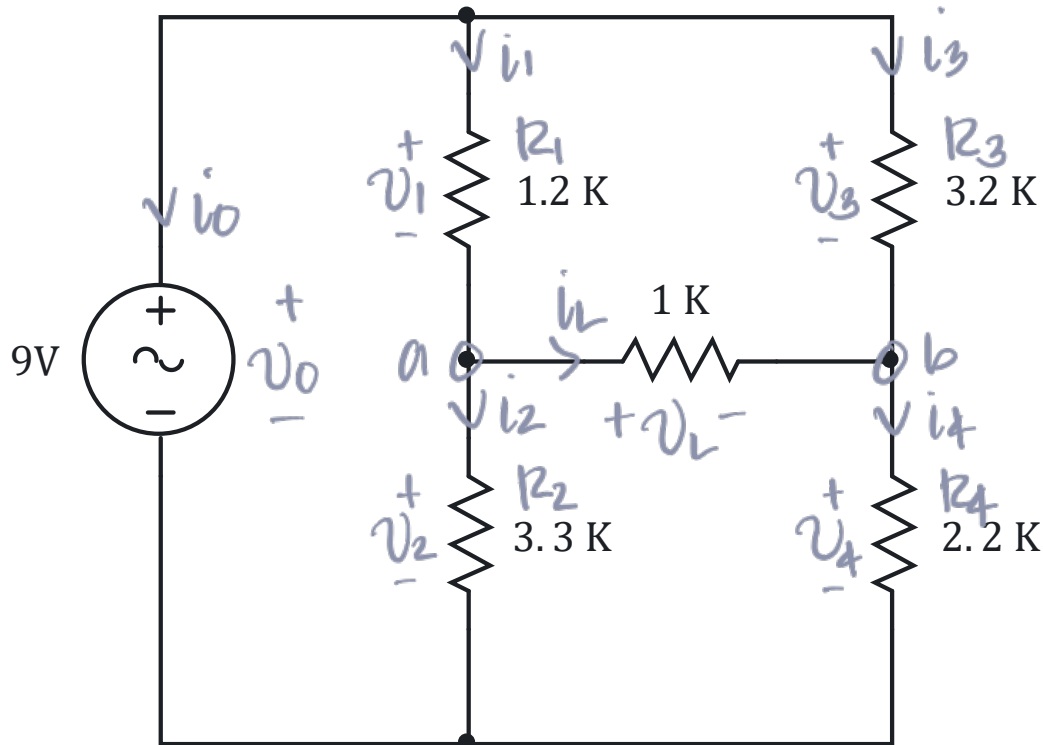
$$P_0 = 153.37 + 100.84 + 67.21$$

$$P_0 = 321.42 \text{ mW}$$

ans

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

KCL @ a

$$-i_1 + i_2 + i_L = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

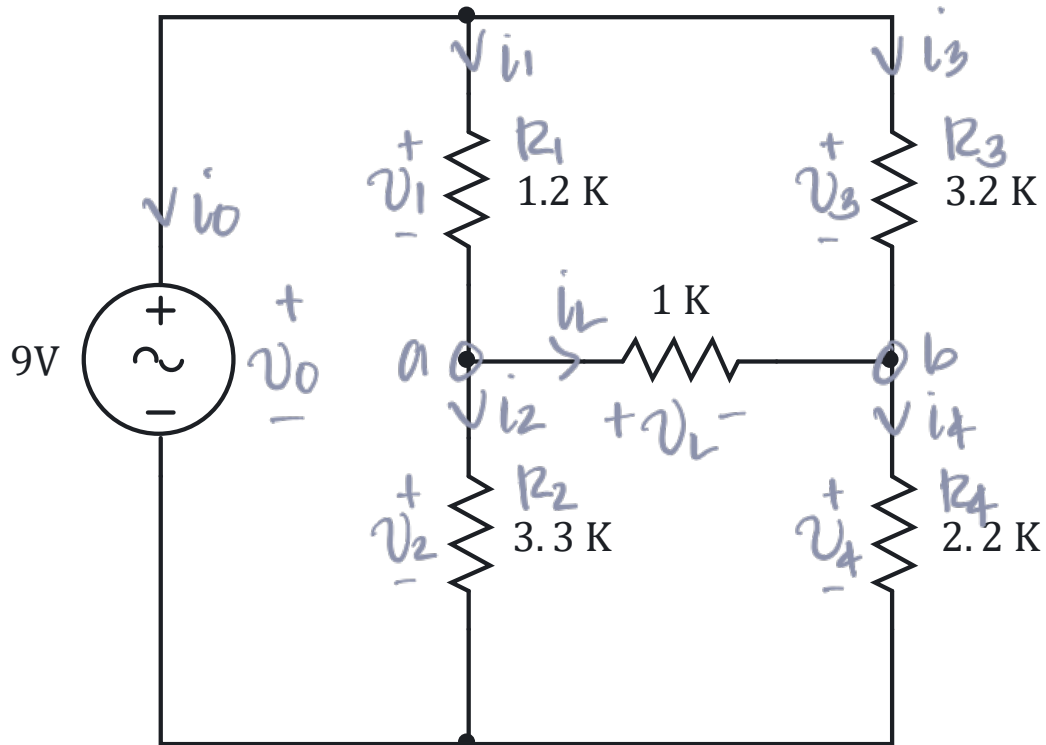
$$-\frac{v_0 - v_a}{R_1} + \frac{v_a}{R_2} + \frac{v_a - v_b}{R_L} = 0$$

$$-v_0 G_1 + v_a G_1 + v_a G_2 + v_a G_L - v_b G_L = 0$$

$$v_a (G_1 + G_2 + G_L) - v_b G_L = v_0 G_1$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$v_a \left(\frac{1}{1.2k} + \frac{1}{3.3k} + \frac{1}{1k} \right) - \frac{1}{1k} v_b = 9 \left(\frac{1}{1.2k} \right)$$

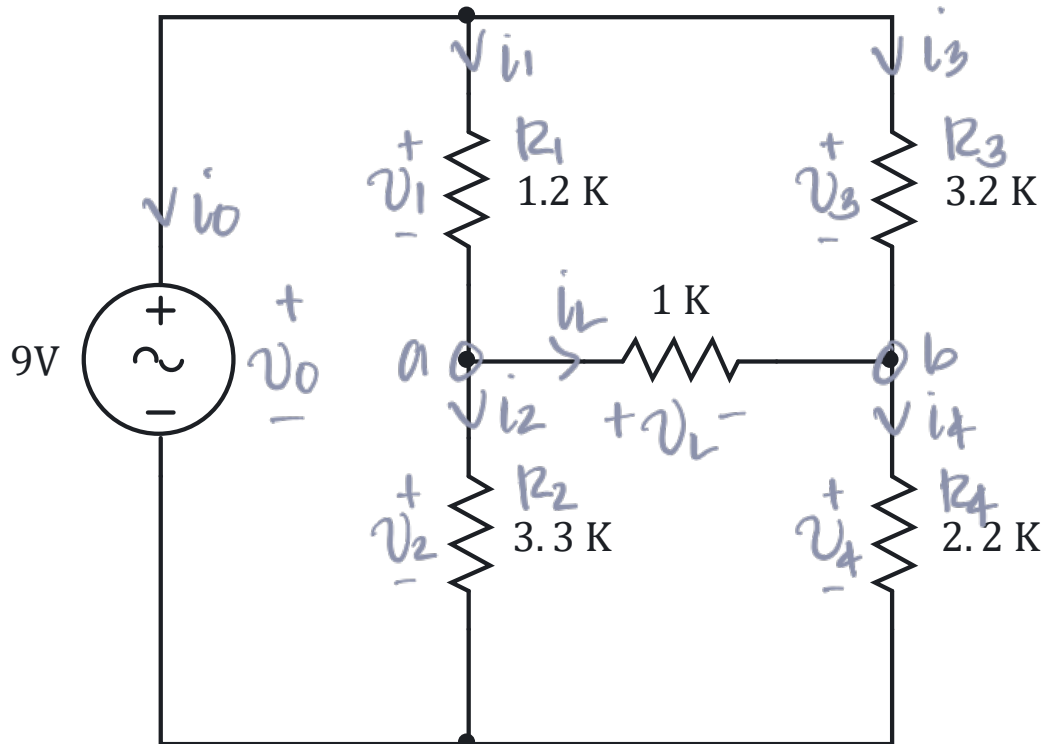
$$1k \left[\frac{47}{22k} v_a - \frac{1}{1k} v_b = \frac{3}{400} \right]$$

$$\frac{47}{22} v_a - v_b = 7.5 \quad (1)$$



EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

KCL @ b

$$-i_3 - i_L + i_4 = 0$$

$$-\frac{v_3}{R_3} - \frac{v_L}{R_L} + \frac{v_4}{R_4} = 0$$

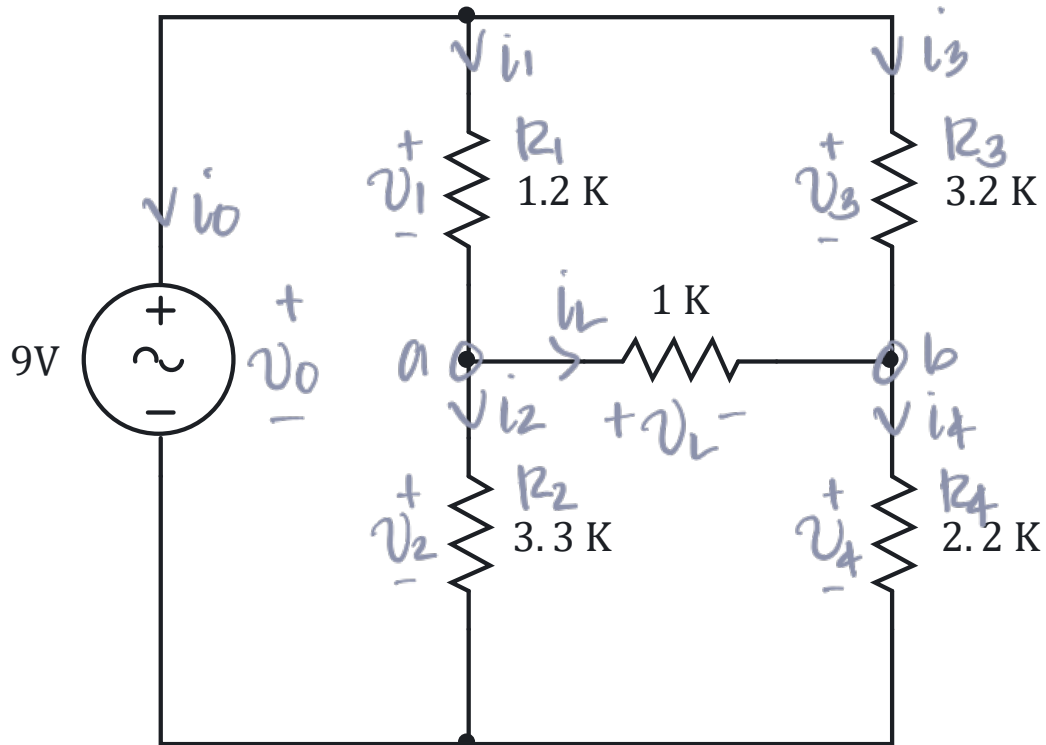
$$-\frac{v_0 - v_b}{R_3} - \frac{v_a - v_b}{R_L} + \frac{v_b}{R_4} = 0$$

$$-v_0 G_3 + v_b G_3 - v_a G_L + v_b G_L + v_b G_4 = 0$$

$$-v_a G_L + v_b (G_3 + G_4 + G_L) = v_0 G_3$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$-\frac{1}{1k}v_a + v_b\left(\frac{1}{3.2k} + \frac{1}{2.2k} + \frac{1}{1k}\right) = 9\left(\frac{1}{3.2k}\right)$$

$$1k \left[-\frac{1}{1k}v_a + \frac{311}{176k}v_b = \frac{9}{3.2k} \right]$$

$$\underline{-v_a + \frac{311}{176}v_b = \frac{9}{3.2} \quad (2)}$$

EXERCISE

Gaussian Elimination

v_a	v_b	
$\frac{47}{22}$	-1	7.6
-1	$\frac{311}{176}$	$\frac{9}{3.2}$

r_1

$$r_2 \leftarrow r_1 + \frac{47}{22} r_2$$

v_a	v_b	
$\frac{47}{22}$	-1	7.6
0	2.78	13.51

r_1

r_2

Solution

$$\frac{47}{22} v_a - v_b = 7.6 \quad (1)$$

$$-v_a + \frac{311}{176} v_b = \frac{9}{3.2} \quad (2)$$



EXERCISE

Gaussian Elimination

V_a	V_b		
$\frac{47}{22}$	-1	7.5	r_1
<u>0</u>	2.78	13.51	r_2

from r_2

$$\frac{2.78 V_b = 13.51}{2.78} \quad \frac{13.51}{2.78}$$

$$\underline{V_b = 4.86 \text{ V}}$$

Solution

from r_1

$$\frac{47}{22} V_a - V_b = 7.5$$

$$\frac{47}{22} V_a = 7.5 + V_b$$

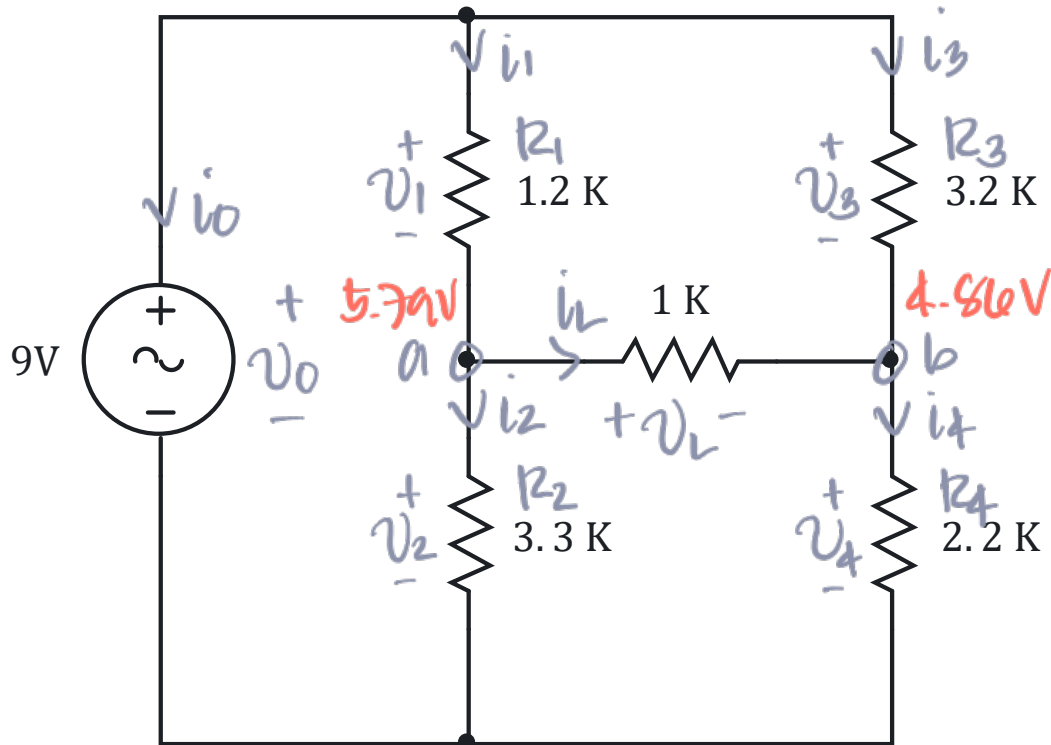
$$V_a = \frac{22(7.5 + \cancel{V_b})}{47} \rightarrow 4.86$$

$$\underline{V_a = 5.79 \text{ V}}$$



EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$V_L = V_a - V_b$$

$$V_L = 5.79 - 4.86$$

$$V_L = 930 \text{ mV}$$

ans

$$i_L = \frac{V_L}{R_L}$$

$$i_L = \frac{930 \text{ mV}}{1 \text{ K}}$$

$$i_L = 930 \mu\text{A}$$

ans

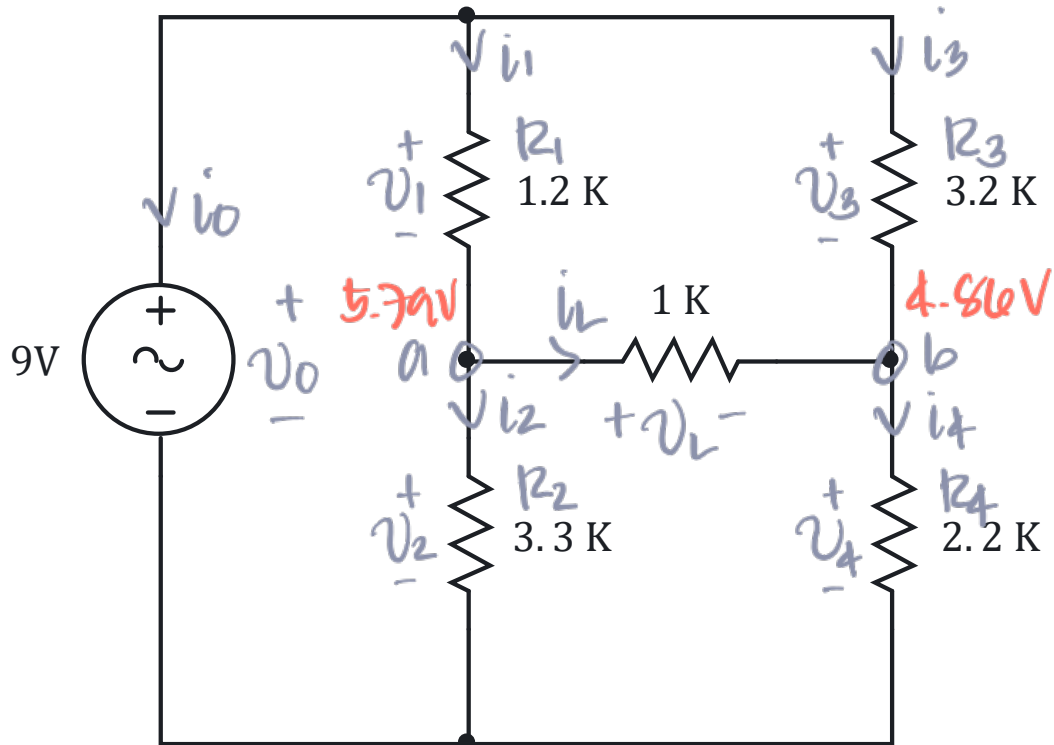
$$P_L = i_L V_L$$

$$P_L = 930 \mu\text{A} (930 \text{ mV})$$

$$P_L = 864.9 \mu\text{W}$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$V_1 = V_0 - V_2$$

$$V_1 = 9 - 5.79$$

$$V_1 = 3.21V$$

ans

$$I_1 = \frac{V_1}{R_1}$$

$$I_1 = \frac{3.21}{1.2K}$$

$$I_1 = 2.68mA$$

ans

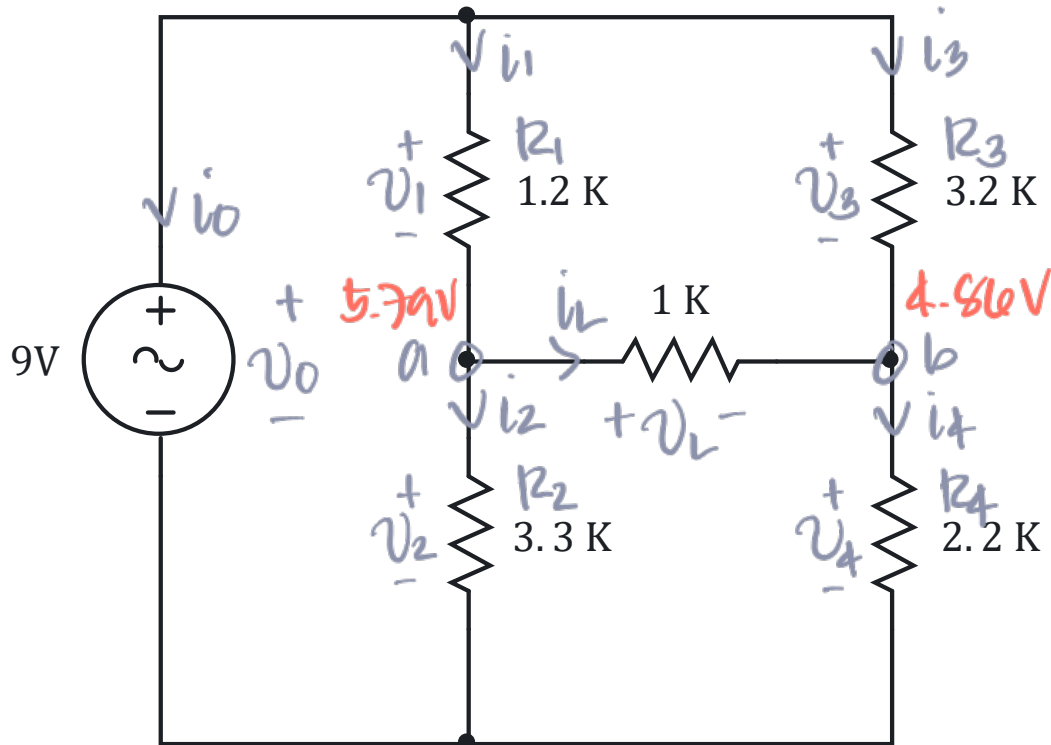
$$P_1 = I_1 V_1$$

$$P_1 = 2.68m(3.21)$$

$$P_1 = 8.60mW$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$V_2 = V_1$$

$$V_2 = 5.79 \text{ V}$$

ans

$$i_2 = \frac{V_2}{R_2}$$

$$i_2 = \frac{5.79}{3.3 \text{ K}}$$

$$i_2 = 1.75 \text{ mA}$$

ans

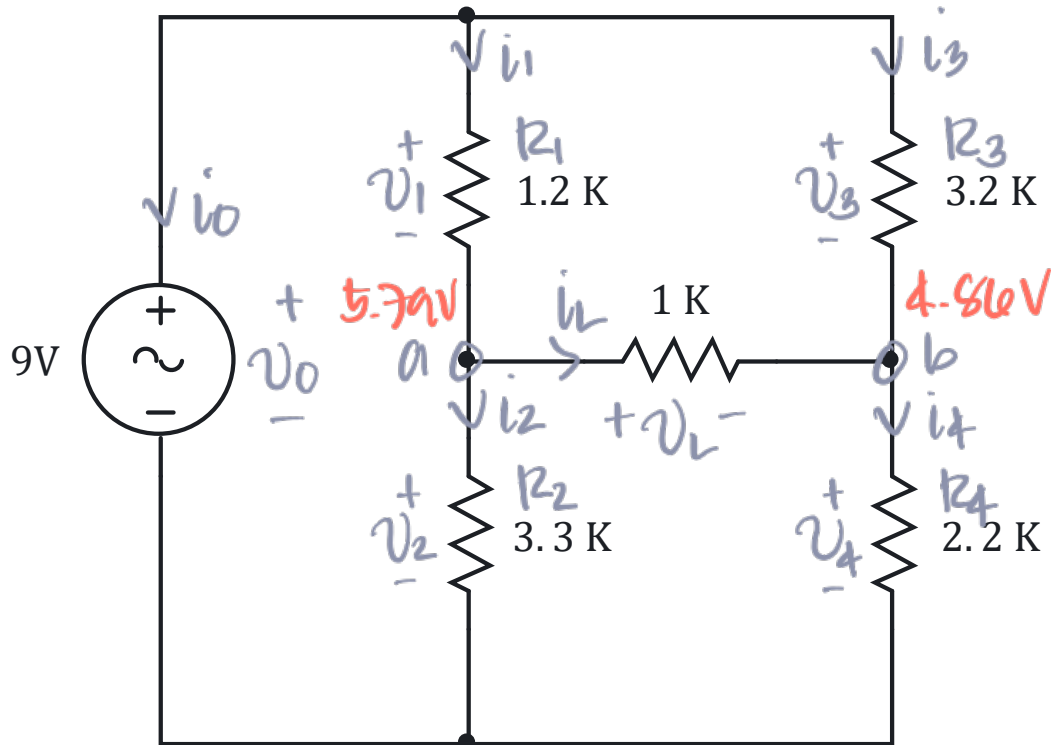
$$P_2 = i_2 V_2$$

$$P_2 = 1.75 \text{ m}(5.79)$$

$$P_2 = 10.13 \text{ mW}$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$V_3 = V_0 - V_2$$

$$V_3 = 9 - 4.86$$

$$V_3 = 4.14 \text{ V}$$

ans

$$i_3 = \frac{V_3}{R_3}$$

$$i_3 = \frac{4.14}{3.2 \text{ K}}$$

$$i_3 = 1.29 \text{ mA}$$

ans

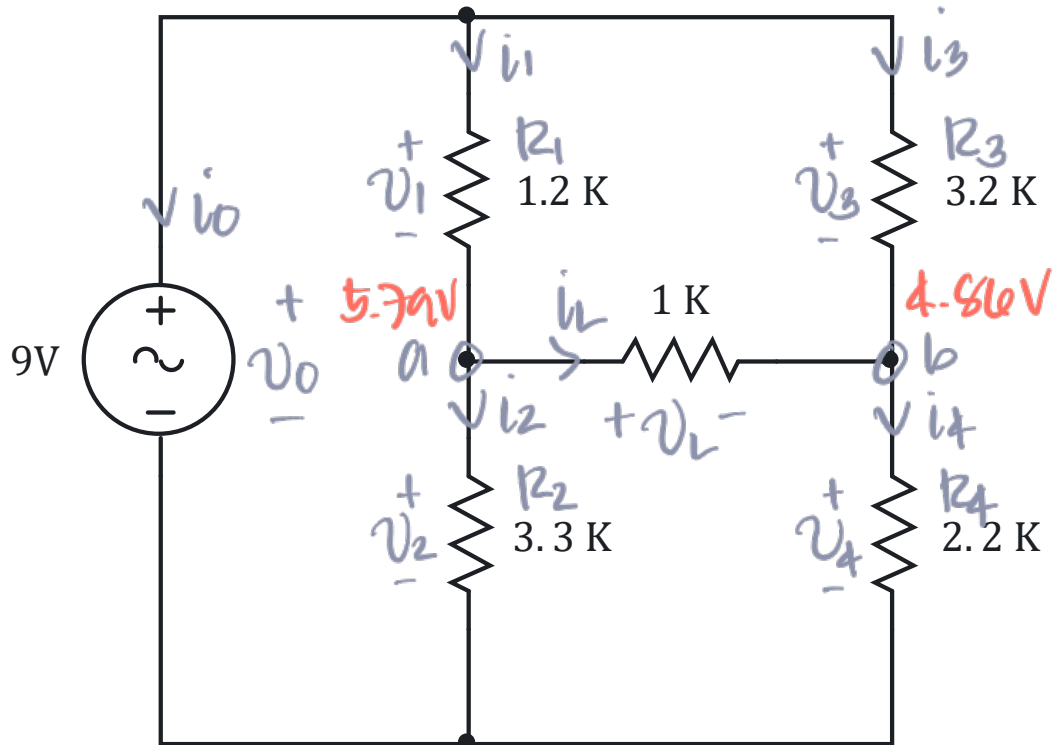
$$P_3 = i_3 V_3$$

$$P_3 = 1.29 \text{ m} (4.14)$$

$$P_3 = 5.34 \text{ mW}$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$V_4 = V_6$$

$$V_4 = 4.86V$$

ans

$$i_4 = \frac{V_4}{R_4}$$

$$i_4 = \frac{4.86}{2.2k}$$

$$i_4 = 2.21mA$$

ans

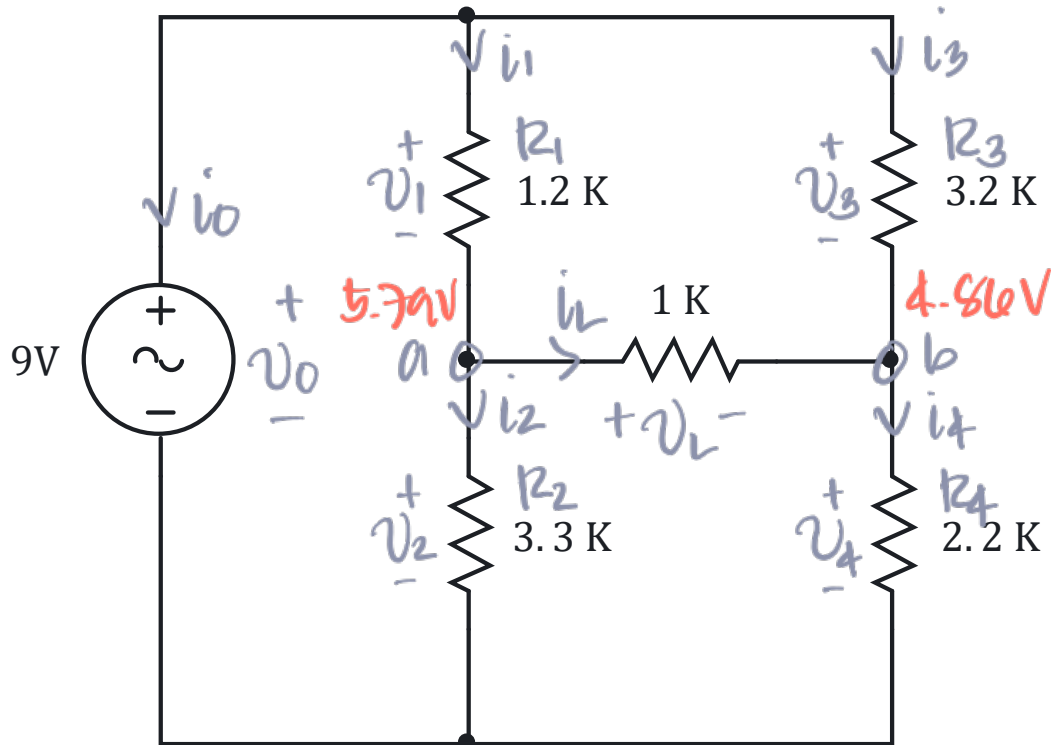
$$P_4 = i_4 V_4$$

$$P_4 = 2.21m(4.86)$$

$$P_4 = 10.74mW$$

EXERCISE

Determine the **current** flowing through each resistor, the **voltage** drop across each resistor, and the total **power** in the given circuit.



Solution

$$P_0 = P_1 + P_2 + P_3 + P_4 + P_L$$

$$P_0 = 8.6m + 10.13m + 5.34m + 10.74m + 864.9m$$

$$P_0 = 35.67mW$$

ans

LABORATORY

